

Layer Composition and Predictive Stability in Transformers: Contextual Reachability and Distributed Head Contributions

Paper 0.5 — Residual / Layer-Composition Research Line

Integrated controlled report — August 2026

Abstract

We study three linked phenomena: how depth changes the stability and correctness of a response under irrelevant input variation, how the self-attention (SA) window limits transport of predictive context, and how heads contribute to stability and class discrimination. The primary instruments are custom causal Transformers trained on known synthetic distributions. A reproduced pilot shows post-block Jensen–Shannon (JS) dispersion $0.0650 \rightarrow 0.0307 \rightarrow 0.0106$ bits despite increasing within-example entropy. A new three-seed matrix generalizes contraction to contiguous n-grams, skip-grams, and binary functors, all at 100% final accuracy under eight nuisances. Dense correctness traces show mean target margin changing from -32.02 to $+10.51$ logits and top-1 fraction from 0 to 1. Absolute margin variance grows from 0 to 1.89, however: correct signal grows much faster than fluctuation rather than fluctuation universally contracting. Stable decisions occur near sublayer depth 3.88 with 0.47-boundary mean settling delay; neither decision depth nor margin SNR supports a scaling-law claim. Local-window experiments show that graph reachability is necessary but not sufficient. In a four-layer/four-head model, equivalent-family head replacement is nearly neutral whereas mismatch is harmful; pair effects are non-additive and motif specificity again fails to predict causal utility ($\rho = -0.0059$). The supported theory is composed, anisotropic, state-dependent predictive refinement—not universal denoising, a sufficient $Lw \geq s$ law, or stable visual head semantics.

1 The question and the three phenomena

Let an input be $x = (g, d)$, where g is the predictive structure and d is constructed nuisance, with

$$Y \perp d \mid g. \quad (1)$$

We ask whether depth organizes the many surface realizations (g, d_i) into a stable predictive class while retaining distinctions between different g . This system-level question is investigated in dependency order:

1. *layer composition*: does stability improve, and is the trajectory monotone?
2. *SA transport*: can decisive context reach the prediction position through a finite window?
3. *heads*: which parallel SA channels stabilize a class or distinguish classes?

Continuous learning is not the experimental objective. Pretrained and prior mechanistic results are retained as secondary external-validity evidence in the appendices.

2 A pedagogical theory of predictive ensembles

For a pre-norm block,

$$r'_\ell = r_\ell + A_\ell(r_\ell), \quad r_{\ell+1} = r'_\ell + M_\ell(r'_\ell). \quad (2)$$

The key point is that layer $\ell + 1$ acts on the state produced by layer ℓ ; layers are composed transformations, not independent estimates. For family g , imagine a cloud

$$\mathcal{E}_\ell(g) = \{r_\ell(g, d_i)\}_{i=1}^m. \quad (3)$$

Its centroid $\mu_{\ell,g}$ describes the common state and covariance $\Sigma_{\ell,g}$ describes realization-sensitive spread. Applying the unembedding at every depth gives distributions $p_{\ell}(\cdot | g, d)$. We measure

$$D_{\text{within}}(\ell) = \mathbb{E}_g \mathbb{E}_d [JS(p_{\ell}(\cdot | g, d), \bar{p}_{\ell,g})], \quad (4)$$

$$D_{\text{between}}(\ell) = \mathbb{E}_{g \neq g'} [JS(\bar{p}_{\ell,g}, \bar{p}_{\ell,g'})], \quad (5)$$

$$R_{\ell} = \frac{D_{\text{between}}(\ell)}{D_{\text{within}}(\ell) + \epsilon}. \quad (6)$$

A useful layer trajectory lowers within-family fluctuation without erasing target signal or between-family separation. Large R_{ℓ} is not self-interpreting when its denominator is nearly zero, so we always report accuracy, target probability, and numerator/denominator with it.

2.1 Correctness-aware order parameters

Let $z_{\ell}(Y)$ be the correct-target logit and let $z_{\ell}(c^*)$ be the strongest competing logit. Define the per-realization margin

$$m_{\ell}(g, d) = z_{\ell}(Y) - z_{\ell}(c^*), \quad (7)$$

which is positive exactly when the target is top-1. Across nuisance realizations, the signal and fluctuation are

$$S_{\ell} = \mathbb{E}_d [m_{\ell}(g, d)], \quad N_{\ell}^2 = \text{Var}_d [m_{\ell}(g, d)], \quad (8)$$

$$\Gamma_{\ell} = \frac{S_{\ell}}{\sqrt{N_{\ell}^2 + \epsilon}}, \quad \Gamma_{\ell}^{\text{robust}} = \frac{\text{median}_d m_{\ell}}{1.4826 \text{MAD}_d(m_{\ell}) + \epsilon}. \quad (9)$$

We always show S_{ℓ} and N_{ℓ}^2 beside Γ_{ℓ} : a large ratio can otherwise arise from a collapsed denominator. Correctness and class organization are complementary: Γ_{ℓ} asks whether the externally specified answer wins reliably, whereas R_{ℓ} asks whether families separate. In representation space we use

$$\chi_{\ell} = \frac{q_{\ell}}{M_{\ell}^{\text{class}} + \epsilon}, \quad q_{\ell} = \mathbb{E}_g \text{Tr} \Sigma_{\ell,g}, \quad M_{\ell}^{\text{class}} = \mathbb{E}_{g \neq g'} \|\mu_{\ell,g} - \mu_{\ell,g'}\|^2. \quad (10)$$

The first top-1 depth is the first boundary where $m_{\ell} > 0$; stable depth is the first boundary after which it remains positive. Never-correct trajectories use a missing sentinel, never the final layer.

For boundary-to-boundary progress $a_{\ell i} = m_{\ell+1,i} - m_{\ell,i}$, write $a_{\ell i} = \mu_{\ell}^{\Delta} + \epsilon_{\ell i}^{\Delta}$. Then

$$S_L^{\text{cum}} = \sum_{\ell} \mu_{\ell}^{\Delta}, \quad (11)$$

$$(N_L^{\text{cum}})^2 = \sum_{\ell} (\sigma_{\ell}^{\Delta})^2 + 2 \sum_{\ell < k} \text{Cov}(\epsilon_{\ell}^{\Delta}, \epsilon_k^{\Delta}). \quad (12)$$

This decomposition makes repair visible: negative cross-layer covariance means that fluctuations introduced at one boundary tend to be counteracted later.

2.2 Why non-monotonicity is expected

For a perturbation δr_{ℓ} , linearize one block:

$$\delta r_{\ell+1} \approx J_{\ell} \delta r_{\ell}, \quad J_{\ell} = I + \frac{\partial F_{\ell}}{\partial r}. \quad (13)$$

After L blocks the relevant map is the product $J_{L-1} \cdots J_0$. One factor may expand a nuisance direction and a later factor may rotate or contract it. Hence final contraction does not imply contraction at every layer. We test the approximation with Jacobian–vector products (JVPs), never by constructing a dense Jacobian.

2.3 Context as a causal cone

With a local window w , one layer can move information at most w positions in the mask graph. Thus $Lw \geq s$ is a necessary reachability condition for a source s positions away in the simple mask used here. It is not sufficient: attention weights, width, optimization, and intermediate coding can still prevent useful transport.

2.4 Heads as vector-field components

Write $A_\ell(r) = \sum_h A_{\ell h}(r)$. For head h , define

$$U_{\ell h}^{\text{stab}} = D_{\text{within}}^{(-h)} - D_{\text{within}}^{\text{intact}}, \quad (14)$$

$$U_{\ell h}^{\text{disc}} = D_{\text{between}}^{\text{intact}} - D_{\text{between}}^{(-h)}. \quad (15)$$

For two heads, $\Gamma(h, k) = U(h, k) - U(h) - U(k)$ distinguishes synergy ($\Gamma > 0$) from redundancy or compensation ($\Gamma < 0$). These are causal, task-conditioned quantities; an attractive attention motif is not a substitute.

3 Controlled methods

The generator records predictive family, held-out surface identity, target, pattern tokens/length/span, nuisance tokens/count/type/seed, answer-changing status, continuation entropy, training frequency, split, and generator seed. Group 1 uses contiguous n-grams, skip-grams, and binary functors: nine families, three model seeds, nuisance counts 0/4/8, twelve realizations, width 48, four layers, and four heads. At the initial pre-SA state and every post-SA/post-block boundary we retain target logit/probability/rank/margin, strongest competitor, top-1 result, entropy, distribution hash, residual covariance, first/stable decision depth, reversals, layer-update signal/noise, and cross-layer covariance. Group 2 uses eight (s, w, L) cells, four predictive classes, an exactly tested local mask, and a separately labelled local-pattern task. Group 3 uses a trained four-layer/four-head full-attention model and evaluates every head with zero ablation, batch-mean replacement, same-family replacement, mismatched replacement, and every within-layer pair. Manifests store resolved configs, seeds, data identities, and checkpoint hashes. Family/seed bootstrap is used for the confirmatory Group 1 contrasts; head scans are descriptive and no row-level significance is asserted.

Part I

I. Layer composition and response stability

4 Pilot reproduction and generalization

The prior fixed-depth result reproduces exactly: under pattern evidence and four distractors, target-probability variance is $0.0347 \rightarrow 0.0106 \rightarrow 0.0109$, JS dispersion is $0.0650 \rightarrow 0.0307 \rightarrow 0.0106$ bits, and within-example entropy is $1.326 \rightarrow 2.094 \rightarrow 2.277$ bits. Stable response across realizations is therefore different from low entropy within one realization.

In the new matrix, all three generators achieve 100% final accuracy with eight nuisances. First-to-last within-JS changes are -0.0244 bits for contiguous n-grams (family/seed CI $[-0.0414, -0.0098]$), -0.0415 for skip-grams ($[-0.0791, -0.0129]$), and -0.0402 for binary functors ($[-0.0738, -0.0143]$). Stability generalizes beyond contiguous n-grams without sacrificing target signal.

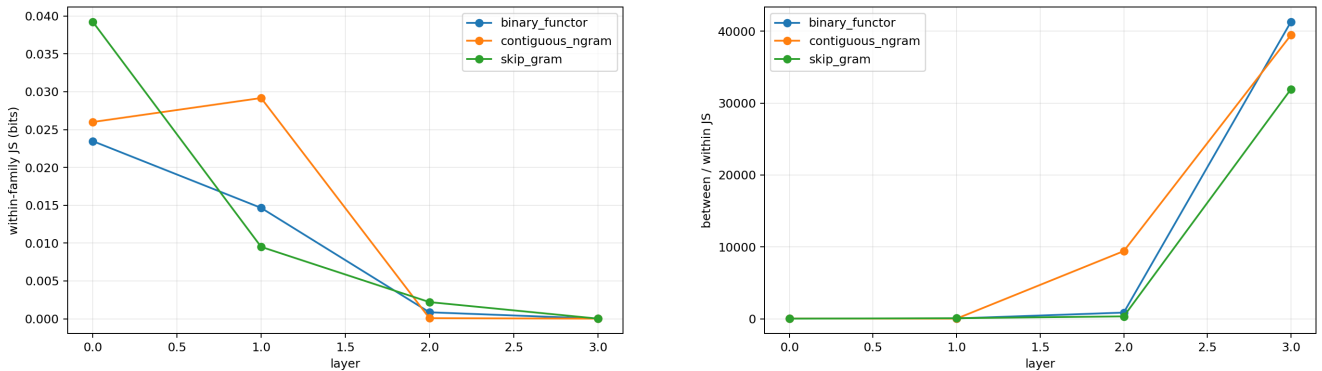


Figure 1: Within-family JS and the separation-to-fluctuation ratio across depth under eight nuisances.

5 Correct signal grows faster than margin fluctuation

At eight nuisances, the aggregate mean correct-target margin changes from -32.02 logits initially to $+10.51$ finally, mean target probability from 1.4×10^{-10} to 0.9998 , and top-1 fraction from 0 to 1 . Each generator independently changes from negative to positive margin. This rules out the “stable but uniformly wrong” failure mode.

The stronger absolute-contraction hypothesis is nevertheless false here. Margin variance starts at zero because the aligned prediction-position state is initially identical across nuisance realizations; it finishes at 1.89 . Final median margin SNR is $+10.24$. Thus reliability comes from roughly 42.5 logits of signal growth outpacing fluctuation growth, not from an SNR denominator collapse. Prediction-space JS can contract even while logit-margin variance expands because the softmax geometry and the mean decision direction both change.

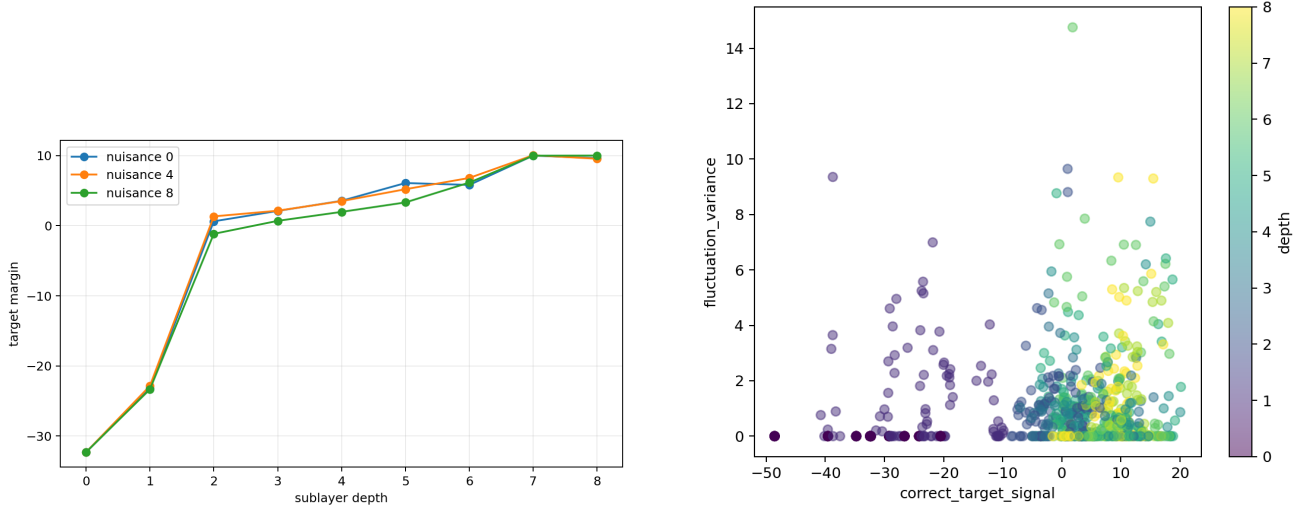


Figure 2: Correct-target margin across sublayer depth and the signal–fluctuation phase plane. The trajectory moves from confidently wrong and identical to confidently correct with finite nuisance variation.

6 Decision formation, update covariance, and scaling non-results

Mean first/stable top-1 depths are $3.00/3.40$ boundaries at nuisance 0, $3.08/3.48$ at nuisance 4, and $3.41/3.88$ at nuisance 8. Mean settling delays are 0.48 , 0.40 , and 0.47 ; mean correctness reversals remain 1.37 – 1.52 . Nuisance modestly delays decision formation, but a five-form comparison finds no scaling law: the best stable-depth fit has $R^2 = 0.013$ and held-out error 1.65 boundaries. Positive-margin SNR fits explain at most 1.0% of variance; logarithmic and linear BIC differ by less than one. A power law is unsupported.

Mean boundary update is $+5.12$ logits and 73.7% of family/seed/noise transition cells have positive mean progress. Median cumulative signal is $+40.39$ logits. Median summed marginal update variance is 3.55 , but median summed off-diagonal covariance is -2.12 , leaving actual cumulative fluctuation variance 0.51 . Later updates therefore cancel a substantial part of earlier realization-specific fluctuation.

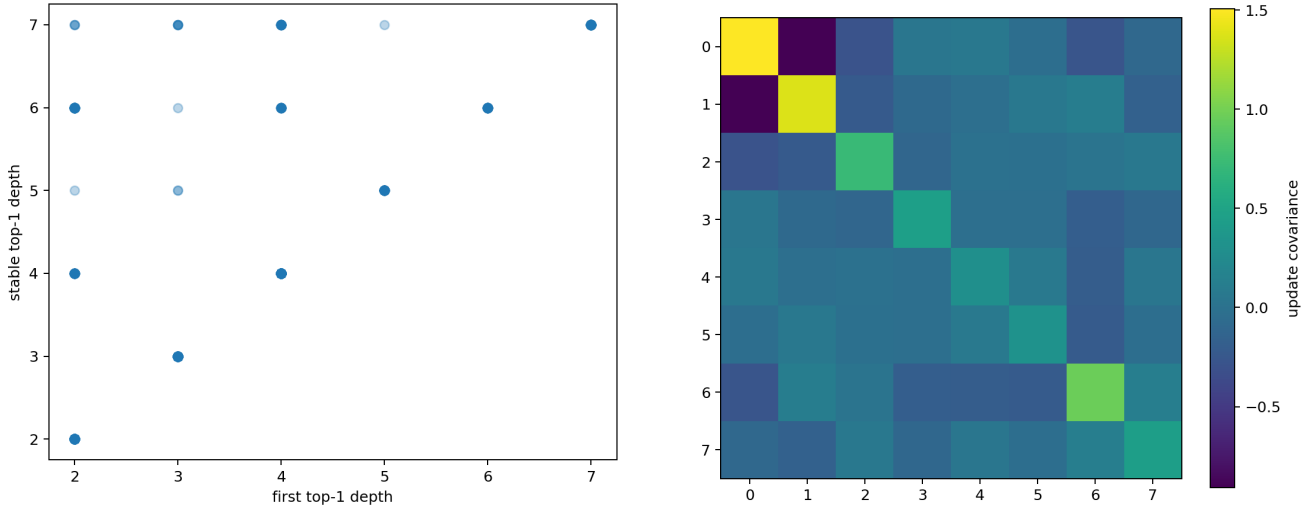


Figure 3: First versus stable decision boundaries and cross-boundary margin-update covariance. Negative off-diagonal structure implements fluctuation cancellation/repair.

Entropy remains independent of correctness: its correlation with aggregate margin is only 0.081. It can increase while margin crosses zero and while within-family JS contracts.

7 Non-monotone composition and local linearization

Only 7/27 family-seed trajectories contract monotonically; 11 expand then contract, five oscillate, three contract late, and one does not contract. This favors cumulative nonlinear composition over a layerwise denoising story.

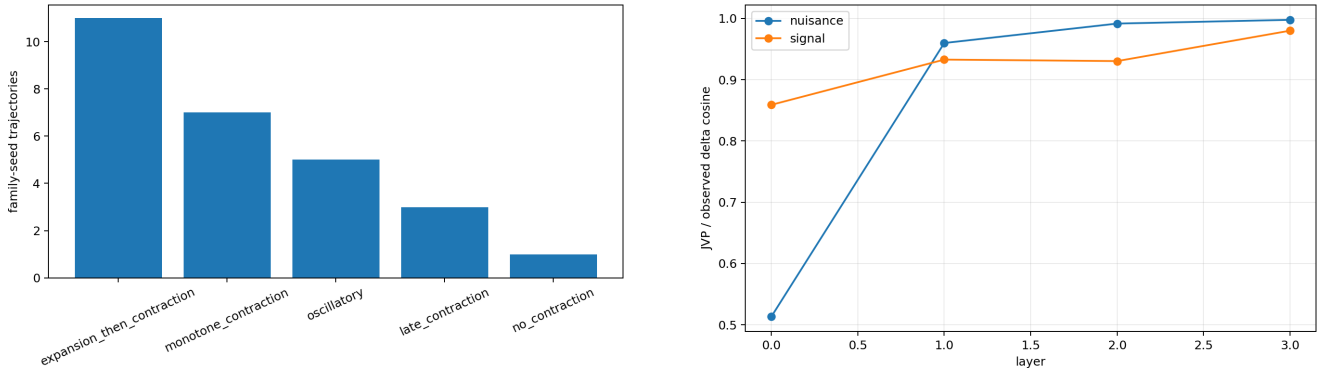


Figure 4: Trajectory classes and local JVP agreement. Median JVP/observed cosine is 0.984 for nuisance and 0.928 for signal directions; prediction-space relative errors are 0.179 and 0.311.

The Jacobian describes small nuisance propagation better than the larger answer-changing displacement, but neither local statistic determines the product of later transformations.

Part II

II. Self-attention window and contextual transport

8 Reachability is necessary, not sufficient

The mask tests verify exact graph reachability. Unreachable long-range cells remain near four-way chance: accuracies are 0.188 for $(s, w, L) = (2, 1, 1)$, 0.234 for $(4, 1, 2)$, and 0.234 for $(8, 2, 2)$. Reachable $(2, 1, 2)$ reaches 0.797; $(4, 1, 4)$, $(8, 4, 2)$, and $(8, \text{full}, 2)$ reach 1.0. Yet reachable $(8, 2, 4)$ reaches only 0.078. The causal cone is a necessary structural bound, not a sufficient learning law.

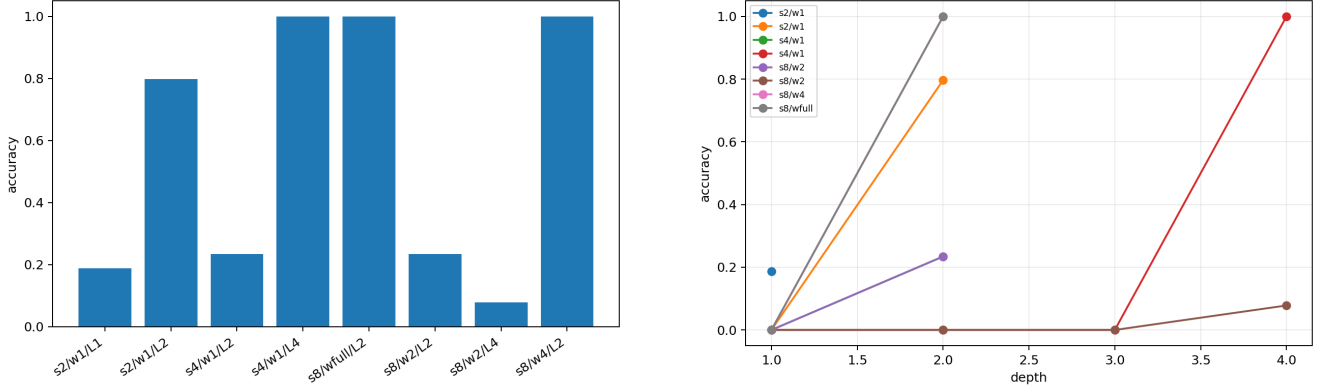


Figure 5: Final accuracy by span/window/depth cell and layerwise functional arrival.

Local-pattern accuracy is 0.75–0.953 in most cells, showing substantial robustness when the decisive token is retained, but it is not uniformly at ceiling. The present smoke run does not independently identify a nested override trajectory; that confirmatory extension remains open.

Part III

III. Heads and the SA contribution

9 Replacement, specialization, and interaction

Across 16 layer/head cells, mean target-log-probability drops are 0.0316 nats for zero ablation, 0.0108 for mean replacement, 0.0006 for equivalent-family replacement, and 0.0194 for mismatched replacement. Zero ablation increases within-family JS by 4.0×10^{-5} bits and reduces between-family JS by 3.93×10^{-4} on average. Mismatch produces the largest discrimination damage (1.44×10^{-3} bits). Stability and discrimination are therefore related but non-identical head roles.

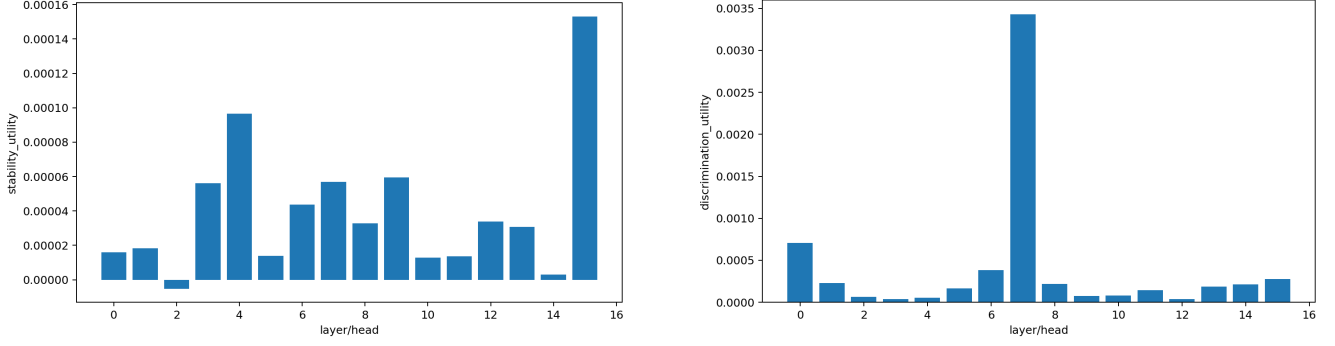


Figure 6: Causal stability and discrimination utilities for all layer/head cells.

Pair interactions range from -0.0116 to $+0.0187$ nats, with median $+0.0050$, so individual utilities do not add. Median correlation of head-output norms is 0.002 , rejecting a simple globally correlated-channel account without establishing independence of the full vectors. Motif specificity is again unrelated to causal utility ($\rho = -0.0059$ across 16 cells), closely reproducing the prior -0.009 null.

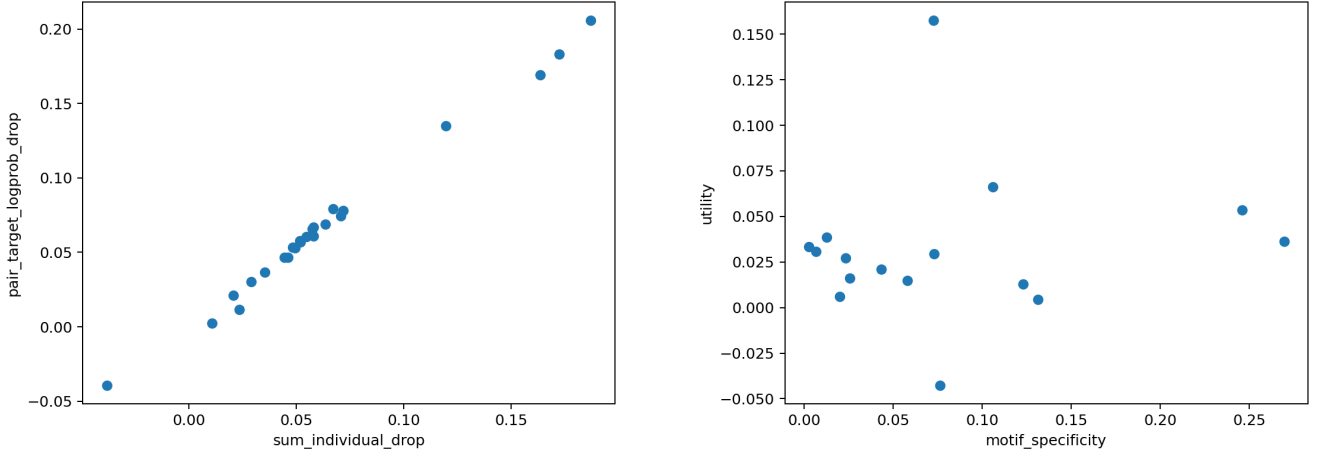


Figure 7: Non-additive pair effects and the repeated motif-causal null.

10 Integrated interpretation

The statistical view is supported in relative form: predictive-family JS contracts, correct margin becomes strongly positive, and signal grows faster than absolute margin fluctuation. The dynamical view is supported descriptively: decisions reverse, trajectories are usually non-monotone, cross-layer fluctuations anticorrelate, and local JVP accuracy differs by perturbation type. The information-theoretic view is only partially supported because class separation is measured but mutual information is not estimated. The statistical-mechanical view is likewise partial: correctness signal, fluctuation, R , χ , and covariance are measured, but neither empirical scaling nor a finite-size phase transition is supported.

The most economical account is *state-dependent anisotropic refinement*. Layers transport, rotate, expand, and contract different perturbations; SA reachability constrains which distinctions can enter the prediction position; heads supply parallel, interacting vector-field components. No single layer or head need be a complete estimator, and final stabilization need not be monotone.

11 Limitations and open tests

The new runs satisfy smoke rather than full scaling gates. Group 1 uses three generators and one architecture scale; the tested range cannot support universal exponents. Zero nuisance has identical realizations and hence exactly

zero fluctuation, producing extreme SNR values that are retained but excluded from positive-margin scaling fits. Group 2 has one seed per cell, does not complete nested override or formal competing-law fits, and contains a reachable optimization failure. Group 3 has one seed/model, no pattern-length recruitment comparison, no multi-seed head-role reproducibility, and only one head-count point; it does not establish head-count scaling. Huge R values arise from near-zero within-family denominators and are not treated as precise effect sizes.

A Prior controlled component and trajectory experiments

The preserved earlier experiment contains 21,630 atlas rows, 1,920 component rows, 1,920 motif/control rows, 2,880 entropy rows, 3,840 equivalence patches, and dense checkpoints. Familiar deterministic mappings are FFN-heavier under zero ablation (0.478 versus 0.301 nats) but not matched replacement; override/repair is SA-heavier (1.497 versus 0.261). Equivalent update replacement changes outputs less than nonequivalent replacement (layer-0 SA JS 0.028 versus 0.436 bits). Diagnostic logit progress often disagrees with causal magnitude, and controlled frequency/mechanism regression has only $R^2 = 0.073$. These results motivate distributed computation but are secondary to the three investigations.

B Pretrained external validity and manifold control

The earlier four-family matrix found equivalent-minus-nonequivalent advantages of 1.047 nats for Pythia step 1,000, 1.206 for final Pythia, and 0.599 for Qwen, but no reliable random-initialization effect. The expanded 32-family matrix found fully-nonequivalent advantages of 2.552 nats for final Pythia (CI [1.991, 3.070]) and 0.763 for Qwen ([0.623, 0.899]), with syntax mismatch more harmful than semantic mismatch. This evidence is sharply qualified by behavioral competence (1.6% for Pythia versus 68.8% for Qwen) and an on-manifold falsification: Qwen’s residual-nearest nonequivalent donor costs -0.022 nats (CI $[-0.082, 0.023]$), essentially identical to equivalent replacement. Rank-1 removal remains harmful (Pythia -1.019 , Qwen -0.837 nats), while addition is harmful. Common subspaces are contributory and state-dependent, but functional labels have not been isolated from residual proximity.

C Override, training, and variance controls

Answer-changing contexts are never labelled nuisance. In the legacy grid their JS contraction is -0.0328 bits (CI $[-0.0543, -0.0141]$), weaker than irrelevant-nuisance contraction -0.0474 ($[-0.0778, -0.0217]$). Random checkpoints have low dispersion only because target signal is absent. Training-time SA/FF trajectories are non-monotone, and stable attention motifs do not predict causal SA utility. These negative controls constrain the theory rather than being discarded.

D Relationship to later continual-learning work

Only here, after the mechanistic results, do we state the downstream implication. Stable predictive equivalence could become a diagnostic precursor for future adaptation, but Paper 0.5 neither implements nor validates continual learning. Common-direction addition is not beneficial, on-manifold controls weaken label-specific transfer, and the separate natural held-out utility gate remains negative. Persistent prototypes or adapters therefore remain blocked pending matched-budget future utility, override preservation, contamination controls, and rollback.

E Reproducibility

```
PYTHONPATH=src python -m cl.experiments.paper05_layer_composition
--group 1 --config configs/paper05/group1.json --device cpu
PYTHONPATH=src python -m cl.experiments.paper05_layer_composition
--group 2 --config configs/paper05/group2.json --device cpu
PYTHONPATH=src python -m cl.experiments.paper05_layer_composition
--group 3 --config configs/paper05/group3.json --device cpu
PYTHONPATH=src python -m cl.analysis.paper05_layer_composition
PYTHONPATH=src python -m cl.experiments.paper05_correctness
```


All prior artifacts remain under `results/`; new raw, aggregate, figure, summary, and manifest artifacts are under `results/three_parts/`. Tests cover deterministic replay, residual identities, trace parity, local-mask correctness, graph reachability, head-contribution shapes, and zero/replacement equivalence.

F Conclusion

Depth reliably transforms strongly wrong initial margins into stable correct predictions across three controlled pattern generators, but usually through non-monotone composition. Correct signal grows faster than nuisance-conditioned fluctuation; absolute margin variance need not contract. Negative cross-layer update covariance reveals cancellation and repair, while the tested decision-depth and SNR scaling laws fail. SA windows impose a causal reachability constraint whose satisfaction does not guarantee learning. Heads contribute in distributed, non-additive ways; equivalent head states transfer better than mismatches, whereas attention motifs remain causally uninformative. Together these findings turn “layer refinement” into a measurable theory of predictive ensembles, causal cones, and interacting residual updates while preserving its failures and boundary conditions.

References

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